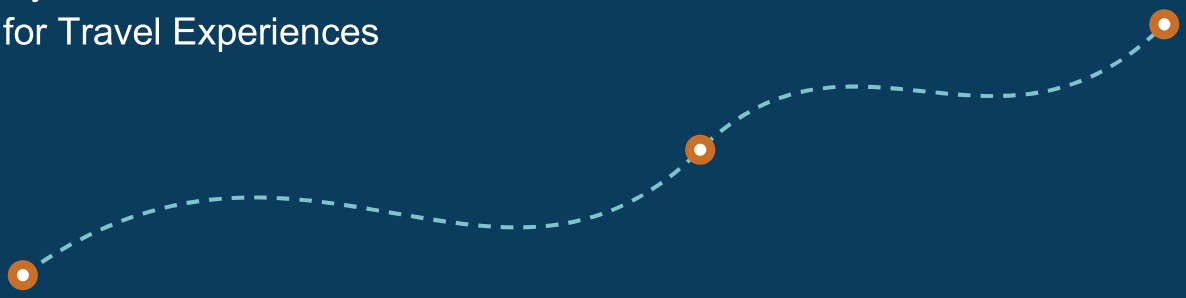
Design Document

Hybrid Retrieval and Recommendation for Travel Experiences



CAPSTONE DESIGN REPORT

Scope: current implementation, kept separate from planned capabilities.

Author: Not specified Version: Not specified Date: Not specified

Source: the uploaded Design_Document.md. No code, datasets, or other attachments were consulted. No evaluation results are reported in this document.

1. Problem Statement and Business Objective

1.1 Problem statement

Travel professionals need to retrieve similar historical travel experiences, compare destinations for a traveler's preferences, and surface highly rated activities, without manually searching raw review records.

1.2 Objective

The service addresses that problem with a workflow that turns each historical record into a searchable case and returns evidence-backed results. It is intended to support three tasks:

- **Retrieve** similar historical travel experiences.
- **Compare** destinations against a traveler's preferences.
- **Surface** highly rated activities.

Business metrics, target users beyond travel professionals, and success criteria are **Not specified** in the source document.

1.3 Scope boundaries

Cost, activity duration, hotel inventory, real-time availability, and live weather are out of scope because they are not present in the data. Recommendation ratings are historical sample signals, not guarantees.

> **Data provenance caveat**
> The added narratives and travel metadata are synthetic or unverified until provenance is confirmed. They must not be described as authentic TripAdvisor reviews.

2. Architecture Design and Data Flow

2.1 Overview

The service loads one enriched CSV row as one travel experience. A normalized document combines destination, travel type, visited activity categories, and review text. A local sentence-transformer creates embeddings stored in Chroma, and BM25 supplies lexical matching. Reciprocal Rank Fusion (RRF) combines the two rankings. A preference classifier extracts supported categories and trip constraints, and deterministic scoring ranks destinations and activities from the retrieved evidence. FastAPI exposes the workflow, Streamlit provides the demonstration UI, and feedback is stored separately in SQLite.

2.2 End-to-end data flow

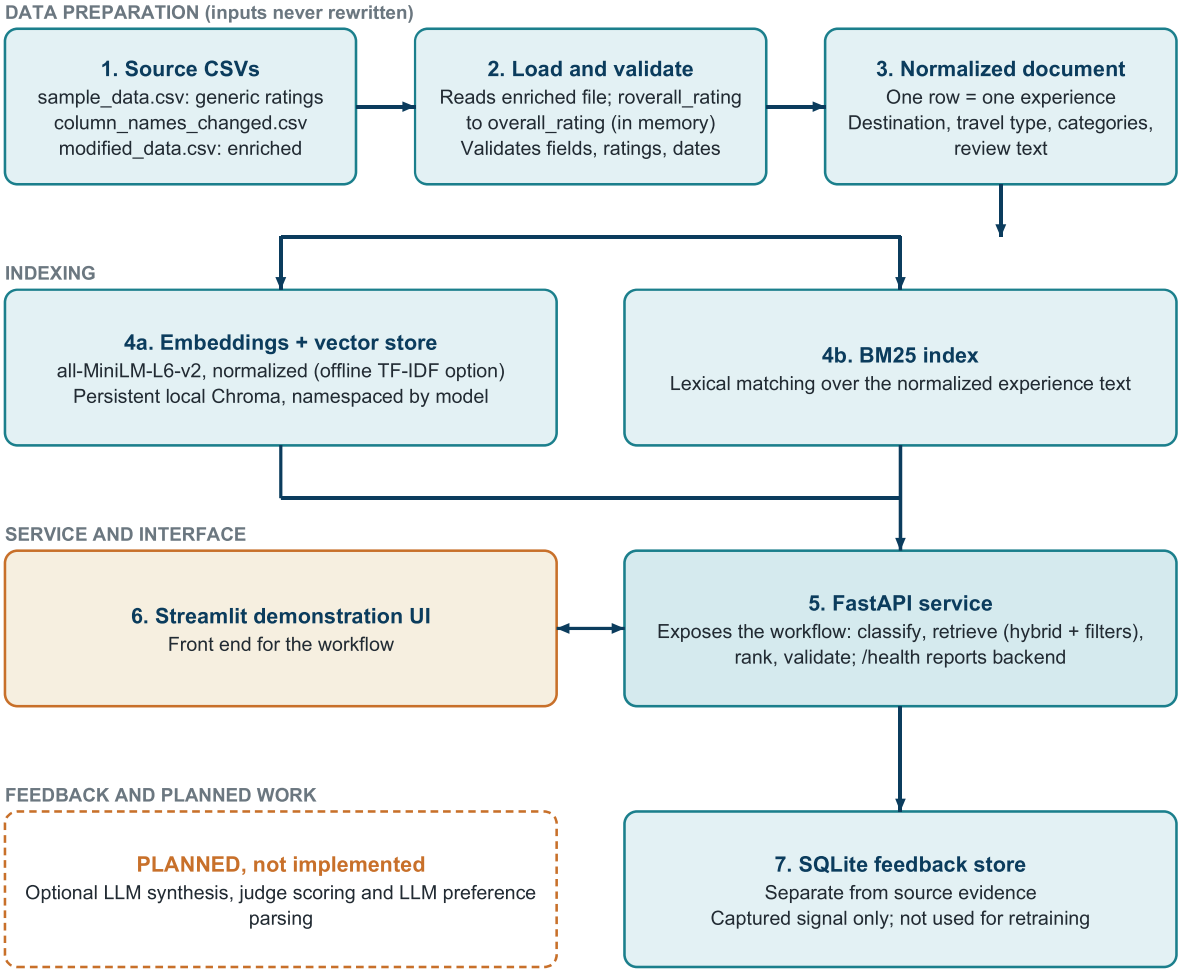


Figure 1. High-level architecture and data flow. Dashed orange outline marks planned work that is not implemented.

2.3 Data contract and preprocessing

File	Role
data/sample_data.csv	Generic source rating matrix.
data/column_names_changed.csv	Renames the 24 category columns of the source matrix.
data/modified_data.csv	Adds destination, categories visited, an overall rating, narrative text, travel type, and visit date.

The application reads the enriched file and normalizes `roverall_rating` to the canonical API field `overall_rating` in memory. It validates required fields, numeric ratings in [0, 5], non-empty metadata, and ISO visit dates. Input CSVs are never rewritten. Price and activity-duration fields are absent, so budget and duration feasibility cannot be computed from evidence.

3. Technology Selection and Rationale

Rationale is stated only where the source document gives one; otherwise it is marked Not specified.

Component	Selection	Rationale (from source)	Status
LLM	Not specified. Optional LLM synthesis, judge scoring, and LLM preference parsing are described.	Not required for current execution. Provider and model: Not specified. Any LLM parsing must keep schema validation and confidence/uncertainty reporting.	**Planned**
Embedding model	sentence-transformers/all-MiniLM-L6-v2 with normalized embeddings; explicit offline TF-IDF option.	Local embeddings avoid mandatory hosted embedding calls but need an initial model download and local storage. TF-IDF is a lower-quality lexical approximation for when model hosting is unavailable; /health reports the active backend.	**Current**
Vector database	Chroma, persistent local collection.	Suitable for a capstone-scale corpus; not intended as a distributed production database. Rebuilt if the corpus fingerprint changes; namespaced by embedding model identity to avoid mixing vector dimensions.	**Current**
Keyword retrieval	BM25 over normalized experience text.	Supplies lexical matching alongside vectors.	**Current**
Rank fusion	Reciprocal Rank Fusion, constant 60.	Combines the vector and BM25 rankings.	**Current**
API framework	FastAPI.	Exposes the workflow. Further rationale: Not specified.	**Current**
User interface	Streamlit.	Demonstration UI. Further rationale: Not specified.	**Current**
Feedback storage	SQLite.	Stores feedback separately from source evidence. Captured signal only; not used for retraining.	**Current**
Evaluation tooling	DeepEval and LLM-as-judge.	Run only with a configured provider. RAGAS: Not specified.	**Planned**

4. Retrieval Strategy

4.1 Document unit and chunking

The unit of retrieval is one complete travel-experience row. There is no within-row chunking, because each row is already a concise case. The normalized text combines destination, travel type, visited activity categories, and review text.

4.2 Embeddings and vector retrieval

Embeddings use `sentence-transformers/all-MiniLM-L6-v2` with normalization and are stored in a persistent local Chroma collection. The collection is rebuilt if the corpus fingerprint changes and is namespaced by embedding model identity. When model hosting is unavailable, an explicit offline TF-IDF vectorizer can be used; the active backend is reported by `/health`.

4.3 Keyword retrieval

BM25 ranks records by lexical match over the normalized experience text. For records with zero keyword matches, the keyword-rank contribution is omitted from fusion.

4.4 Hybrid search and fusion

Reciprocal Rank Fusion with constant 60 combines the vector and BM25 rankings. Metadata filters constrain destination, travel type, and overall-rating minimum and maximum.

4.5 Evidence and ranking

Retrieval returns stable row IDs, source text, metadata, and both component ranks so results can be audited. A deterministic preference classifier extracts supported categories and trip constraints. Deterministic scoring then ranks destinations and activities using the retrieved evidence. The scoring formula, the number of results returned (k), and score weights are **Not specified.**

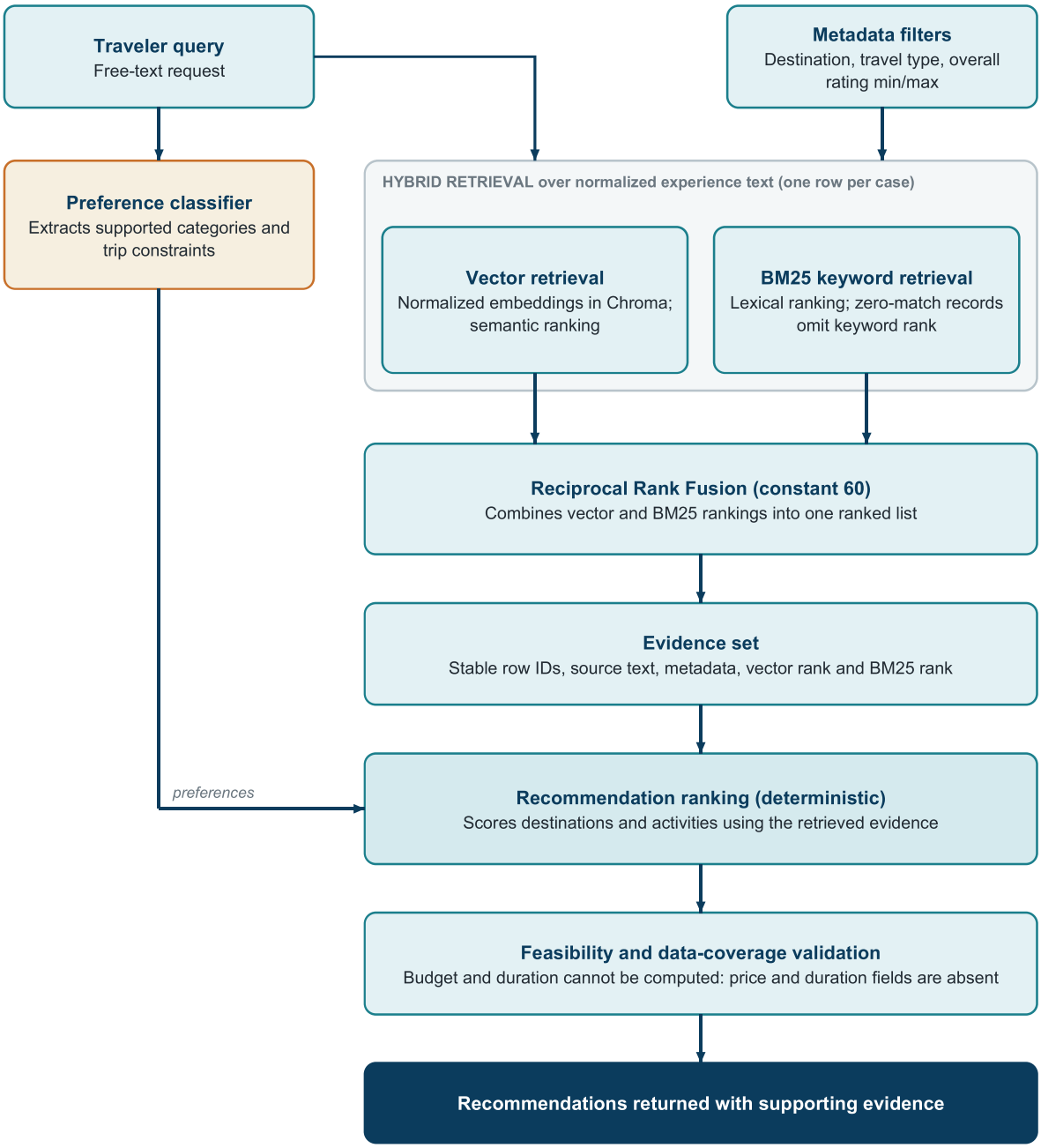


Figure 2. Retrieval workflow from query processing to recommendations. The source does not state the exact stage at which filters are applied; they are shown as an input to retrieval.

5. Agent and Role Design

The current workflow uses deterministic, in-process roles. A coordinator function composes the roles; there is no distributed agent runtime. Optional LLM synthesis and judge scoring are planned and are not required for current execution.

5.1 Roles

Role	Responsibility	Status
Preference classification	Extracts supported categories and trip constraints. Deterministic and basic; optional LLM parsing must retain schema validation and confidence/uncertainty reporting.	**Current**
Evidence retrieval	Hybrid BM25 and vector retrieval with RRF and metadata filters; returns auditable evidence.	**Current**
Recommendation ranking	Deterministic scoring of destinations and activities from retrieved evidence.	**Current**
Feasibility and data-coverage validation	Checks results against what the data can support; budget and duration cannot be computed.	**Current**
LLM synthesis / judge scoring	Optional generation and judging.	**Planned**

5.2 Workflow and communication

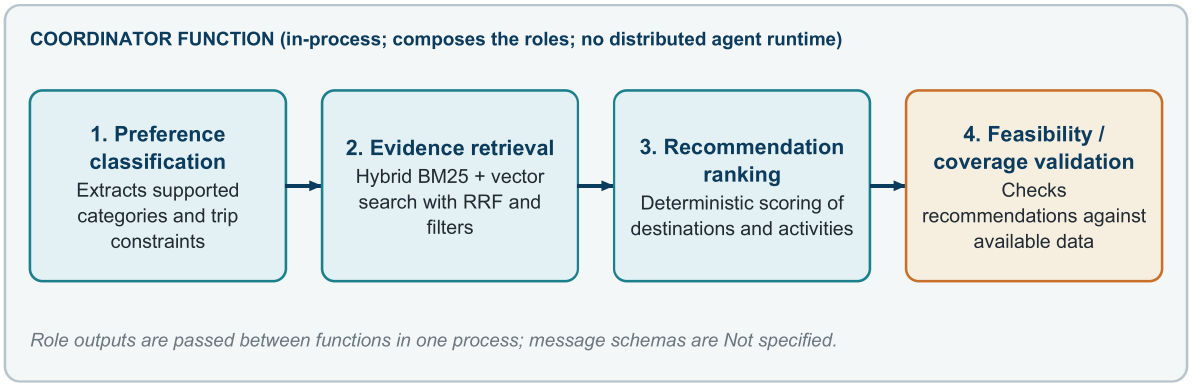


Figure 3. Role workflow composed by a coordinator function. Roles run in a single process.

Roles communicate through function composition inside one process; no message broker or distributed runtime is described. Role-level message schemas, retry behavior, and error handling are **Not specified.**

6. Evaluation Strategy

> **Status**
> No evaluation scores or benchmark results are reported in this document. DeepEval and LLM-as-judge are planned, not run. RAGAS is Not specified in the source.

6.1 Current verification

Current tests cover the following areas. The number of tests and pass rates are Not specified.

- CSV contract validation
- Duration and budget parsing
- Hybrid retrieval
- Metadata filters
- Ranking evidence
- API schemas and endpoints
- Feedback writes

6.2 Planned retrieval evaluation

The next evaluation stage should add a held-out query set with expected destinations and categories, and report the following metrics. Query count and expected labels are Not specified.

Metric	Purpose	Status
Precision@k	Share of the top k results that are relevant.	**Planned**
Recall@k	Share of relevant items found in the top k results.	**Planned**
nDCG	Ranking quality that rewards placing relevant items higher.	**Planned**

6.3 DeepEval, RAGAS, and LLM-as-judge

- **DeepEval and LLM-as-judge:** planned. They should run only with a configured provider and report prompt, model, and version alongside results. Scores must never be invented when unavailable.
- **RAGAS:** Not specified in the source document; no plan or result is stated.
- **Test cases for these frameworks:** Not specified.

7. Challenges, Trade-offs, and Limitations

Area	Trade-off or limitation
Embeddings	Local embeddings avoid mandatory hosted calls but require an initial model download and local storage. The offline TF-IDF option is a lower-quality lexical approximation.
Vector store	Chroma is a persistent index suitable for a capstone-scale corpus, not a distributed production database.
Dataset	The dataset is too small and too enriched to establish general destination quality. Ratings are historical sample signals, not guarantees.
Data provenance	Added narratives and travel metadata are synthetic or unverified until provenance is confirmed; they are not authentic TripAdvisor reviews.
Out-of-scope data	Cost, activity duration, hotel inventory, real-time availability, and live weather are not in the data, so budget and duration feasibility cannot be computed.
Feedback	SQLite feedback is a captured signal only. It is not used for retraining and should not silently overwrite source evidence.
Preference classifier	Currently basic and deterministic. Optional LLM parsing must retain schema validation and confidence/uncertainty reporting.
Evaluation	No held-out evaluation has been reported; DeepEval and LLM-as-judge need a configured provider.

Appendix A. Current versus Planned

Capability	Status
Hybrid retrieval (BM25 + vector, RRF, metadata filters)	**Current**
Deterministic preference classification and ranking	**Current**
FastAPI service, Streamlit UI, SQLite feedback capture	**Current**
Tests for contract, retrieval, filters, ranking evidence, API, feedback	**Current**
Held-out query set with Precision@k, Recall@k, nDCG	**Planned**
DeepEval and LLM-as-judge scoring	**Planned**
Optional LLM synthesis and LLM preference parsing	**Planned**
RAGAS	**Not specified**
LLM provider and model	**Not specified**